

REPLICATION

Environmental Regulations and Air Pollution in India: A Reexamination

Olexiy Kyrychenko

Nijmegen School of Management, Radboud University, Nijmegen, Gelderland, The Netherlands

Correspondence: Olexiy Kyrychenko (olexiy.kyrychenko@ru.nl)

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ABSTRACT

This paper reexamines the effectiveness of environmental regulations in India, originally evaluated by Greenstone and Hanna (2014) using ground-based monitoring data. Replacing this sparse and inconsistent data with satellite-based reanalysis data reveals contrasting air pollution trends and notable differences in the evaluation of policies' effectiveness. The findings underscore the importance of reliable data for the accurate assessment of policy outcomes in regions with limited monitoring infrastructure.

JEL Classification: Q53, Q58, O13, O53

1 | Introduction

This paper reexamines the effectiveness of air pollution control policies in India,¹ building upon the prominent study by Greenstone and Hanna (2014, henceforth GH). Analyzing data from 140 cities during 1987–2007, GH found that mandated catalytic converter policy (CAT) was strongly associated with declines in suspended particulate matter (SPM) and sulfur dioxide (SO₂), while the Supreme Court Action Plans (SCAP) led to a modest reduction in nitrogen dioxide (NO₂).

GH's findings rely on ground-based monitoring data that faced significant limitations during their study period. India's air quality monitoring network was nascent at that time, with few stations, uneven geographic coverage, and reliance on manual monitoring, which often resulted in incomplete and inconsistent data. These shortcomings likely biased GH's reported trends, including their finding of declining air pollution levels, which contrasts with evidence of worsening air quality in India. The prominence of GH's study and these limitations motivate a reexamination using alternative data.

This paper leverages satellite-based reanalysis air pollution data, unavailable to GH, to reexamine the link between environmental regulations and air quality. By integrating observational measurements with numerical models, these data provide consistent estimates where ground-based monitoring is sparse or unreliable. Applying GH's methodology, this analysis finds their results highly sensitive to alternative air pollution measures, offering limited empirical support for the effectiveness of India's air pollution control policies. It also underscores the value of satellite-based reanalysis data for evaluating environmental regulations, particularly in contexts with limited monitoring infrastructure, such as in developing countries.

2 | Data

2.1 | GH's Data Limitations

GH's air pollution data came from India's Central Pollution Control Board (CPCB), which manages a national network of

ground-based monitoring stations. These data have notable limitations that likely compromised its accuracy and reliability for assessing air pollution trends.²

First, the CPCB's monitoring network had limited spatial coverage and temporal consistency during the GH study period. India's monitoring infrastructure was still in infancy, with the number of monitored cities fluctuating from 20 in 1987 to 115 in 2007 likely due to such factors as malfunctioning monitors, relocation, and inconsistent operations.³ Monitors were concentrated in a few severely polluted cities, biasing the data toward these areas and overlooking broader urban pollution patterns across the country (CSE 2023). Manual monitoring and irregular sampling frequencies further contributed to inconsistencies (CPCB 2003; CSE 2023). Many stations in GH's dataset failed to meet the target of 104 annual observations, and some recorded as few as 20 days per year, leading to potentially substantial gaps and inaccuracies in the dataset.

Second, even within monitored cities, the sparse and uneven distribution of air pollution monitors raises concerns about the accuracy and representativeness of city-level data. In 2007, 18% of the sample cities lacked SPM monitors entirely, while over half relied on two or fewer monitors, which may fail to capture significant spatial variability due to factors like emission sources and wind patterns (Sullivan and Krupnick 2018). GH's use of monitor readings simple averages to calculate city-level concentrations may inadequately capture this spatial variability likely misrepresenting pollution trends.⁴ GH also imposed additional restrictions on their sample, further narrowing the analysis based on the availability of air pollution data.⁵

Finally, systemic issues in India's air quality monitoring during GH's study period likely compromised data accuracy. The CPCB's guidelines describe its data as "indicative rather than absolute" (CPCB n.d.), likely reflecting limited quality control, inconsistent calibration, and potential data manipulation.⁶ The lack of data homogenization further exacerbates these issues (Pant et al. 2019).⁷ Alang and Aggarwal (2022) report a 30% discrepancy in PM_{2.5} levels in Varanasi under similar conditions found in two studies. These challenges, coupled with management issues across multiple agencies (CPCB 2003), highlight the need for caution when using CPCB data from this period.

2.2 | Alternative Air Pollution Outcomes

To address issues with CPCB's air pollution data, I leverage publicly available NASA's MERRA-2 reanalysis data. MERRA-2 combines satellite measurements of aerosol optical depth (AOD),⁸ ground-based readings, and sophisticated modeling to provide global gridded estimates for atmospheric and aerosol variables starting from 1980.⁹ I replace CPCB's SPM with MERRA-2 PM_{2.5} estimates, a finer fraction of SPM.¹⁰ Data on PM_{2.5} were unavailable to GH, as its monitoring in India began only in 2009, while MERRA-2 is the only alternative source for these data during their study period.¹¹ NO₂ data were not available for this study within the required temporal and geographic scope.

When using datasets like MERRA-2, it is important to address two key concerns: the implications of spatiotemporal smoothing and the effects of prediction errors.

MERRA-2 data benefit from an assimilation process that integrates observational data with atmospheric models, ensuring consistency and reducing gaps in coverage. However, spatial smoothing due to its grid resolution aggregates data over areas approximately 50 km × 62 km at the equator, potentially obscuring fine-scale air pollution variation. Temporal smoothing similarly averages data across fixed assimilation windows, typically a few hours. These characteristics, while limiting sensitivity to localized and short-term changes, are advantageous for capturing long-term trends and large-scale patterns. Given this study's focus on annual city-level changes in PM_{2.5} and SO₂ concentrations, MERRA-2's strengths align well with the analysis, minimizing potential limitations from spatial and temporal aggregation.¹²

Validation studies indicate that MERRA-2 tends to underestimate PM_{2.5} levels during high-pollution periods due to missing nitrate aerosols and conservative emission estimates.^{13,14} This issue not only is unique to MERRA-2 but also applies to other AOD-based PM_{2.5} datasets, such as those by Di et al. (2016) and van Donkelaar et al. (2019). Fowlie et al. (2019), Greenstone et al. (2021), and Currie et al. (2023) similarly highlight discrepancies between satellite-based and ground-based PM_{2.5} levels, noting challenges related to spatial coverage, uncertainties, and deviations at extreme pollution levels. Despite these prediction errors, MERRA-2 PM_{2.5} reliably captures spatial and temporal trends over larger areas and longer periods, making it suitable for studies like mine and GH's. MERRA-2 data have also been used in economics research, linking PM_{2.5} to reduced economic activity (Dechezleprêtre et al. 2019), manufacturing gains (Fu et al. 2021), and outmigration (Chen et al. 2022). By overcoming the limitations of inconsistent CPCB data, MERRA-2 data allow for a robust reexamination of GH's findings, though its underestimation of PM_{2.5} suggests my results likely represent a lower bound of regulatory effects.

I obtained gridded data for monthly averages of SO₂ and PM_{2.5} components at a 0.5° × 0.625° spatial resolution from NASA's Data and Information Services Center. While SO₂ concentrations were directly available, PM_{2.5} concentrations were calculated using its components—dust (DUST_{2.5}), sea salt (SS_{2.5}), black carbon (BC), organic carbon (OC), and sulfates (SO₄)¹⁵—by applying Equation (1) from Buchard et al. (2016). Annual averages were calculated by aggregating monthly grid cell values. Figure 1 illustrates the spatial distribution of the resulting MERRA-2 PM_{2.5} and SO₂, averaged at the grid level over the study period.

$$\text{PM}_{2.5} = \text{DUST}_{2.5} + \text{SS}_{2.5} + \text{BC} + 1.4 \times \text{OC} + 1.375 \times \text{SO}_4 \quad (1)$$

I then resampled annual data to a finer spatial resolution, reprojected it to a uniform coordinate reference system, and linked it to cities' administrative boundaries, which were created by combining digital maps with the Census definition of urban areas (Census of India, 2011). Finally, I calculated city-level PM_{2.5} and SO₂ by spatially averaging annual grid-level values of all

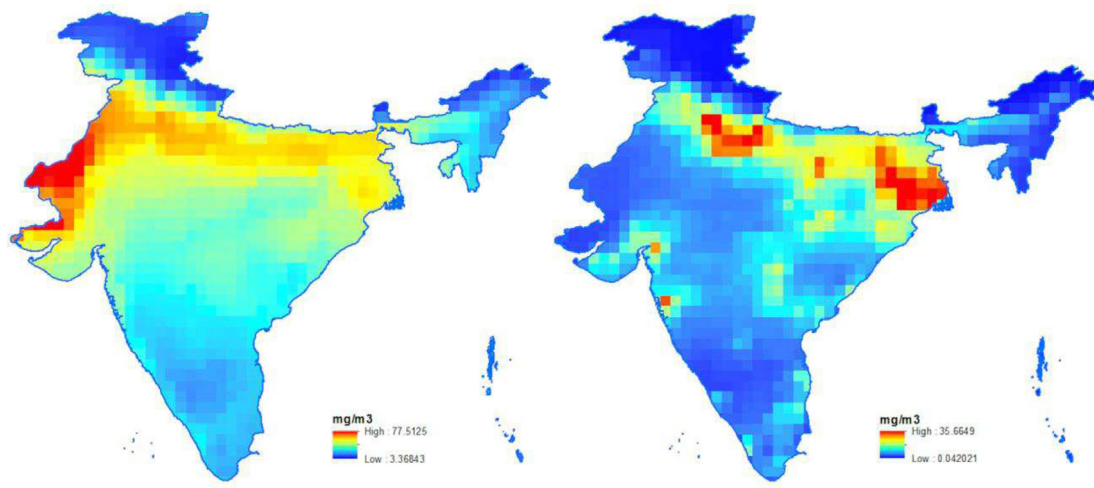
A. Average PM_{2.5}, 1987–2007B. Average SO₂, 1987–2007

FIGURE 1 | Spatial distribution of air pollution concentrations in India, 1987–2007. Note: The figure maps spatial distributions of PM_{2.5} and SO₂ concentrations constructed using MERRA-2 reanalysis data. Panels A and B show long-run average PM_{2.5} and SO₂ concentrations in micrograms per cubic meter for 1987–2007, respectively. Shades of red color depict higher concentrations of the specific air pollutants.

cells within the cities' boundaries.¹⁶ This workflow, implemented using ArcPy scripts in ArcGIS, follows established spatial analysis practices.¹⁷ The final dataset represents city-level annual mean concentrations from 1987 to 2007.

3 | Comparisons of Trends

Figure 2 compares air pollution trends from GH using CPCB data with alternative outcomes based on MERRA-2 data across three data-sample combinations. GH data/GH sample replicates GH's original data and a restricted sample of cities. New data/GH sample combines the revised air pollution outcomes with the same city count as in GH. Finally, new data/new sample leverages the full potential of the revised outcomes, substantially increasing the sample of city-by-year observations to 2940, compared to 1370 and 1344 for GH's particulate matter and SO₂, respectively.

The figure reveals contrasting trends between GH's and the revised air pollution outcomes. While GH's data show declines in SPM and SO₂, the revised outcomes indicate a continuous increase. Appendix Table 1 shows GH's SPM dropped by 17%, from 252.13 µg/m³ in 1987–1990 to 209.42 µg/m³ in 2004–2007. SO₂ remained stable until the late 1990s before falling sharply, from 19.36 µg/m³ in 1987–1990 to 12.19 µg/m³ in 2004–2007, an overall 37% reduction.

MERRA-2 PM_{2.5} for GH's sample rose by 58%, from 22.39 to 35.4 µg/m³ in 2004–2007 compared to 1987–1990. Similarly, MERRA-2 SO₂ increased by 42%, from 4.98 to 7.08 µg/m³. The increases are even more pronounced for the extended sample of cities, with PM_{2.5} rising by 67% and SO₂ by 89%. Notably, the trends between the new data/GH sample and new data/new sample are strikingly similar, despite the larger sample size of the latter. This suggests the revised air pollution outcomes, rather than sample size, drive the observed trends.

The differing trends between GH's SPM and MERRA-2 PM_{2.5} cannot be explained by the fact that SPM and PM_{2.5} are not directly comparable. Using SPM-to-PM₁₀ and PM₁₀-to-PM_{2.5} ratios from Nilekani (2014) and Greenstone et al. (2015), I convert GH's SPM into PM_{2.5}, referred to as GH PM_{2.5}. Column 2 of Appendix Table 1 provides summary statistics for GH PM_{2.5}, which reveal qualitatively similar trend differences between GH's SPM/PM_{2.5} and MERRA-2 PM_{2.5}.

Dramatic differences in air pollution trends can likely be attributed to the limitations of GH's data, stemming from the CPCB's sparse and inconsistent monitoring network. These shortcomings may explain the erratic year-to-year patterns in GH's SPM/PM_{2.5} data (left-hand graph in panel A of Figure 2). In contrast, MERRA-2 reanalysis data, consistently compiled throughout GH's study period, likely offer a more reliable measure of air pollution trends.

Although direct ground truth for comparison is unavailable for GH's study period, indirect evidence supports the validity of MERRA-2 PM_{2.5}.¹⁸ MERRA-2 trends align with major economic events, such as the 1987 Black Monday crash and the 1991 Indian economic crisis, both linked to declines in air pollution. The improvement in air quality during 2005–2006, attributed by GH to intensified environmental regulation enforcement, also mirrors these patterns. Overall, MERRA-2 PM_{2.5} trends reflect findings from studies linking economic downturns to temporary air pollution reductions (e.g., Chay and Greenstone 2003; Sanders 2012; Kyrychenko 2024) and align with evidence of worsening air quality in India and China (Greenstone et al. 2015; Ebenstein et al. 2015; Fu et al. 2021). Moreover, EDGAR PM_{2.5} and SO₂ emissions and MERRA-2 PM_{2.5} and SO₂ concentrations reveal strikingly similar trends, with substantial increases over time and a pronounced spike shortly after 2000 (Appendix Figure 1). This alignment between two independent data sources

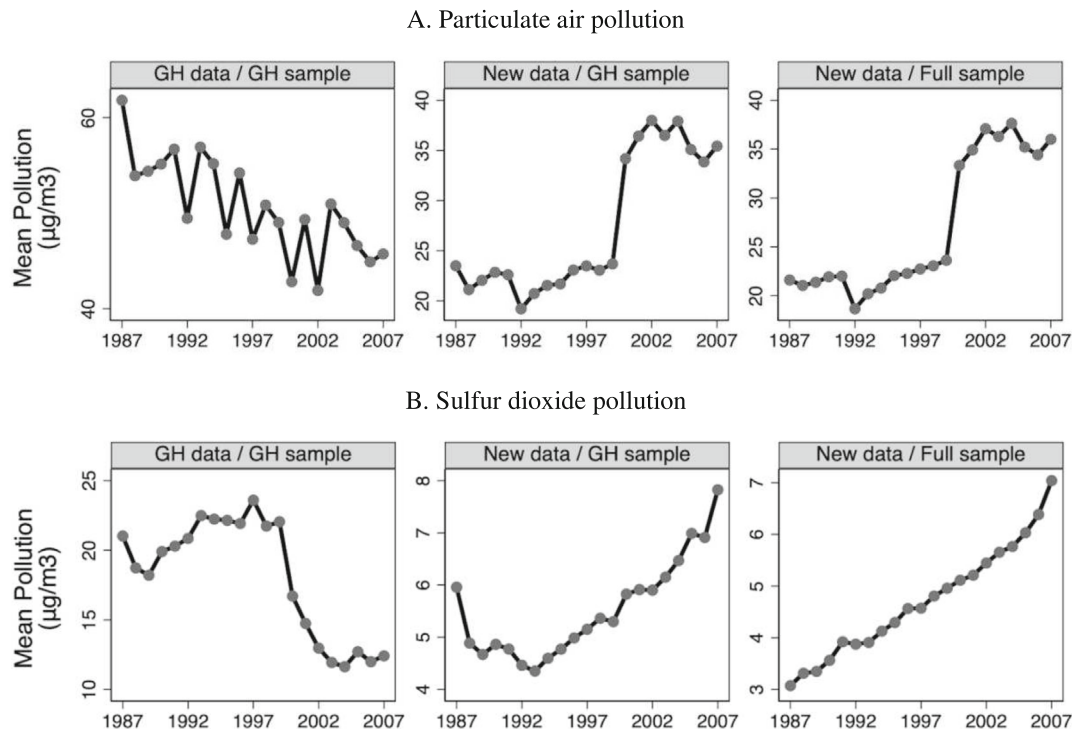


FIGURE 2 | Trends in air pollution, 1987–2007. Note: The figure plots annual city-level average concentrations of particulate air pollution (Panel A) and SO_2 (Panel B) across three data-sample combinations. Left-hand graphs show $\text{PM}_{2.5}$ and SO_2 trends in GH's data for their restricted sample of cities. GH's $\text{PM}_{2.5}$ is obtained by converting from original GH's SPM using $\text{SPM}-\text{PM}_{10}-\text{PM}_{2.5}$ ratios, where PM_{10} is particulate matter with a diameter less than $10\text{ }\mu\text{m}$. Right-hand graphs show trends in $\text{PM}_{2.5}$ and SO_2 estimates for the full sample of 140 cities, while the middle graphs plot the trends for the same pollutants for GH's sample of cities. GH's air pollution data were drawn from the CPCB ground-based monitoring network, while the revised air pollution data — from the MERRA-2 satellite-based reanalysis.

further validates the reliability of MERRA-2 $\text{PM}_{2.5}$ and SO_2 data and supports their use in evaluating long-term air quality trends during periods where ground-based measurements are unreliable.

The sharp increase in MERRA-2 $\text{PM}_{2.5}$ around 2000 may raise concerns about data accuracy, but closer examination supports its validity. Appendix Figure 2 attributes this rise primarily to an increase in $\text{DUST}_{2.5}$, likely caused by dust storms (Prasad and Singh 2007), along with simultaneous increases in SO_4 , organic carbon, and black carbon—byproducts of industrial activities, vehicular emissions, and biomass burning. Similar patterns have been observed in Chen et al. (2022) for China and Kumar et al. (2023) for India, with the latter linking India's $\text{PM}_{2.5}$ increase to economic liberalization and rapid industrialization. Regional variation in trends further supports MERRA-2 $\text{PM}_{2.5}$ accuracy: not all cities experienced the same sharp increase, indicating that the data capture localized environmental changes rather than systematic bias.¹⁹

The comparison reveals significant disparities in particulate and SO_2 air pollution trends between CPCB and MERRA-2 data. This underscores the need to reexamine the effectiveness of environmental regulations using alternative air pollution outcomes, which may produce findings distinct from those in GH.

4 | Effects of Alternative Air Pollution Outcomes

In this section, I test the sensitivity of GH's findings to the alternative air pollution outcomes by following their empirical strategy, which combines event study and difference-in-differences designs in a two-step approach (Equations (2) and (3)).

$$Y_{ct} = \alpha + \sum_{\tau} \sigma_{\tau} D_{\tau,ct} + \beta X_{ct} + \mu_t + \gamma_c + \epsilon_{ct} \quad (2)$$

where Y_{ct} represents air pollutant concentrations in city c in year t . $D_{\tau,ct}$ is a vector of indicators for each year before and after a policy's adoption, with $\tau = 0$ in the enactment year, ranging from -17 to $+12$, and set to zero for nonadopting cities. X_{ct} includes controls (consumption per capita and literacy rates). γ_c and μ_t denote city and year fixed effects. Equation (2) is weighted by district-urban population. The coefficients σ_{τ} measures annual average outcomes in pre- and postadoption. The estimated coefficients $\hat{\sigma}_{\tau}$ are then used in Equation (3) that corresponds to equation (2C) in GH.

$$\hat{\sigma}_{\tau} = \pi_0 + \pi_1 1(\text{Policy})_{\tau} + \pi_2 \tau + \pi_3 (1(\text{Policy})_{\tau} \cdot \tau) + \epsilon_{\tau} \quad (3)$$

where $1(\text{Policy})_{\tau}$ is a dummy variable indicating that the policy is in force; τ is a linear time trend controlling for the differential pre-existing trends in adopting cities. $1(\text{Policy})_{\tau} \cdot \tau$ allows the policy effects to evolve over time. ϵ_{τ} is the heteroskedasticity-consistent standard errors. GH weigh Equation (3) by the inverse of the

standard errors of the relevant σ_τ to account for differences in precision in the σ_τ 's estimation. This specification tests for policy impacts after adjusting for trends in the outcome variable (π_2), allowing for both a mean shift (π_1) and a trend break (π_3). From this equation, GH report policy effects 5 years after implementation, $\pi_1 + 5\pi_3$. Additionally, they complement the two-step approach by its numerically identical one-step version.²⁰

By adhering precisely to GH's methodology, I ensure that any disparities in the results stem solely from differences in air pollution data. Table 1 shows the estimated impacts of the SCAP and CAT policies on $PM_{2.5}$ and SO_2 , constructed using CPCB and MERRA-2 data. For each policy-pollutant and data-sample combination, the table reports estimates from GH's second-step specification (Equation (3)) and its one-step analog. The table focuses

on the policies' impact 5 years after implementation, which GH considered as the key indicator of effectiveness.

Columns 1–2 replicate GH's results. The outcome variables in these columns are the city-level annual average $PM_{2.5}$ and SO_2 , where $PM_{2.5}$ is an indicator of particulate air pollution converted from GH's SPM using $SPM-PM_{10}-PM_{2.5}$ ratios. Appendix Table 2 compares replication results using GH's SPM and $PM_{2.5}$. The results are qualitatively similar in terms of the sign and statistical significance of the coefficients. Relying on this comparison, I use GH's $PM_{2.5}$ in the rest of the analysis. I successfully reproduce GH's results, confirming that the CAT policy is strongly associated with the reduction in $PM_{2.5}$ and SO_2 5 years after the policy implementation by 10.75 $\mu g/m^3$ and 13.45 $\mu g/m^3$ or 19% and 69% of the 1987–1990 nationwide mean concentrations.

TABLE 1 | Effects of alternative air pollution outcomes.

	Replication		Reexamination		Reexamination	
	GH data/GH sample		New data/GH sample		New data/Full sample	
	Two-step (1)	One-step (2)	Two-step (3)	One-step (4)	Two-step (5)	One-step (6)
<i>Supreme Court Action Plans</i>						
<i>Panel A. $PM_{2.5}$</i>						
5-year effect	−0.05	0.20	8.56*	9.65**	9.09**	9.11*
<i>p</i> -value	[0.99]	[0.98]	[0.06]	[0.05]	[0.02]	[0.06]
Observations	11	1165	11	1165	11	2720
1987–1990 mean		55.8		22.39		21.5
<i>Panel B. SO_2</i>						
5-year effect	−1.74	−0.78	0.13	0.18	0.40	0.38
<i>p</i> -value	[0.21]	[0.87]	[0.62]	[0.56]	[0.18]	[0.19]
Observations	11	1158	11	1158	11	2720
1987–1990 mean		19.36		4.98		3.33
<i>Mandated catalytic converters</i>						
<i>Panel C. $PM_{2.5}$</i>						
5-year effect	−10.75**	−10.71*	−2.36	−1.88	−1.93	−1.71
<i>p</i> -value	[0.04]	[0.06]	[0.25]	[0.20]	[0.18]	[0.12]
Observations	17	1165	17	1165	17	2720
1987–1990 mean		55.8		22.39		21.5
<i>Panel D. SO_2</i>						
5-year effect	−13.45***	−12.69**	0.93***	0.92***	0.24*	0.24*
<i>p</i> -value	[0.00]	[0.02]	[0.00]	[0.00]	[0.08]	[0.07]
Observations	17	1158	17	1158	17	2720
1987–1990 mean		19.36		4.98		3.33

Note: The table tests the sensitivity of GH's findings to revised air pollution outcomes. It reports estimates from fitting GH's second-step equation, odd columns, and its one-step version, even columns, for the effects of SCAP and CAT policies on $PM_{2.5}$ (Panels A and C) and SO_2 (Panels B and D) 5 years after implementation. Columns 1–2 use GH's original data to replicate their results. I substitute GH's SPM by GH's $PM_{2.5}$ converted from GH's SPM using $SPM-PM_{10}-PM_{2.5}$ ratios for comparability with the policies' effects on MERRA-2 $PM_{2.5}$. Columns 3–4 exploit the same sample of cities as in GH and revised $PM_{2.5}$ and SO_2 outcomes to reexamine GH findings. Columns 5–6 reexamine GH results by taking full advantage of the revised outcomes and fitting GH's second-step equation and its one-step version to all available city-by-year observations. Five-year effect is an estimate of the policies' effect 5 years after implementation. *p*-value of hypothesis tests for the significance of 5-year effects is reported below the estimates in square brackets. The table also shows $PM_{2.5}$ and SO_2 nationwide mean concentrations during 1987–1990.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Columns 3–4 use the same sample of cities as GH but replace the original air pollution outcomes with MERRA-2 $PM_{2.5}$ and SO_2 . The effects of this substitution are quantified by the differences between the coefficients in columns 1–2 and 3–4 (e.g., column 1–column 3). The alternative air pollution outcomes result in substantial changes in the effects of the policies. In contrast to GH, the significance of the CAT's impact on $PM_{2.5}$ 5 years after implementation vanish. While the CAT's effects on SO_2 remain significant, the coefficients become positive and small. The SCAP's effects on $PM_{2.5}$ also differ sharply from GH's findings, turning positive, large, and significant.²¹ Another change is that the SCAP's effects on SO_2 become positive, small, and insignificant.

Finally, columns 5–6 take full advantage of MERRA-2 air pollution data, reporting coefficients estimated from GH's specifications with the alternative air pollution outcomes and extended observations. The differences between columns 3–4 and 5–6 (e.g., column 3 - column 5) capture the effects of the expanded sample. The SCAP and CAT's effects on $PM_{2.5}$ and SO_2 5 years after implementation change moderately compared to columns 3–4. While the general pattern of the impacts remains consistent, the magnitudes differ. The most notable changes are in the policies' effects on SO_2 . Additionally, the coefficients from GH's two-step and one-step approaches in columns 5–6 become more similar, likely due to the larger sample size.

The analysis highlights the high sensitivity of GH's findings to alternative air pollution outcomes, emphasizing the limitations of the CPCB data discussed in Section 2.1. GH's results largely do not hold when their original outcomes are replaced with data from alternative sources. Policies identified as strongly linked to air quality improvements in GH's study exhibit substantial changes in effect size, significance, and direction. These discrepancies likely reflect measurement errors stemming from deficiencies in the ground-based monitoring network. The CPCB's limitations, including incomplete and uneven spatial coverage, irregular and manual sampling, and inconsistent data quality, appear to have biased GH's conclusions. In contrast, the satellite-based reanalysis data provide limited empirical support for the effectiveness of India's SCAP and CAT policies in reducing air pollution, calling into question the true impact of these environmental regulations.

5 | Conclusions

This paper reexamines the effectiveness of environmental regulations in India, originally assessed by Greenstone and Hanna (2014). Using GH's empirical strategy but replacing ground-based air pollution data with satellite-based reanalysis data, this study uncovers two key findings. First, air pollution trends differ significantly between the two data sources: while GH reported declines, the revised data show a continuous increase. Second, GH's findings are highly sensitive to this change—policies linked to air quality improvements in GH appear ineffective when evaluated with reanalysis data. These findings highlight the importance of reliable data in assessing environmental regulations, particularly in regions with limited monitoring infrastructure.

Greenstone and Hanna's (2014) study remains a key contribution to understanding environmental regulations in developing countries, where severe air pollution and data scarcity pose major challenges. This study builds on their work by demonstrating how reanalysis data can complement and improve traditional monitoring approaches. The findings suggest that leveraging satellite-based estimates enhances our understanding of policy effectiveness, providing a more comprehensive view of air quality trends in regions with weak institutions and sparse ground-based data. By advancing the empirical methodology for studying air pollution, this paper highlights the potential of satellite-based reanalysis data to fill gaps in evaluating environmental regulations, supporting more informed policy design and implementation.

Acknowledgments

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Data Availability Statement

The data and code that support the findings of this study are hosted by the ZBW Journal Data Archive and are accessible via the following DOI: <http://dx.doi.org/10.15456/jae.2025057.1625081642>.

Endnotes

- ¹ GH also examines water pollution regulations, but this study focuses on their analysis of air pollution regulations.
- ² For more details on the limitations of the CPCB's ground-based monitoring data, see Appendix 1.
- ³ Despite the large number of monitoring stations in India, its network is inadequate for effective air quality management. Brauer et al. (2019) found that India's monitor density of ~0.14 monitors/million people is far lower than other populous countries like China (1.2) and the United States (3.4). Even in the United States, the EPA's monitors cover 63% of the population in less than 20% of counties, yet are still considered sparse (Sullivan and Krupnick 2018; Fowlie et al. 2019). This highlights the challenge of achieving comprehensive air quality coverage, even in well-resourced networks.
- ⁴ Ideally, spatial interpolation methods, such as Kriging or inverse distance weighting, would better address this issue. These methods account for spatial variability, providing a more accurate way to aggregate monitor data into citywide concentrations. Without them, aggregation may yield misleading estimates, especially in cities with uneven monitoring coverage.
- ⁵ Policy-adopting cities were included if they had at least one observation three or more years before and four or more years after policy implementation. Non-adopting cities and adopting cities without post-policy data were included if they had at least two readings.
- ⁶ There is evidence that local officials in developing countries may manipulate ground-based air pollution readings (Andrews 2008; Chen

et al. 2012; Ghanem and Zhang 2014). This manipulation includes strategically placing monitors in less polluted areas of cities or relocating monitors to locations upwind to decrease local pollution concentrations.

⁷ For more details about the lack of homogenization of CPCB monitor data, see Appendix 2.

⁸ Aerosol optical depth measures the extent of light extinction caused by airborne particles. It is a critical input for estimating ground-level particulate matter pollution, as it directly reflects the presence and concentration of aerosols in the atmosphere.

⁹ For more details about MERRA-2 and the process of reanalysis, see Appendix 3.

¹⁰ PM_{2.5} refers to particles smaller than 2.5 µm, whereas SPM includes particles smaller than 100 µm. An increasing number of social scientists focus on PM_{2.5} to study the effectiveness of environmental regulations, health effects, and the economic impacts of pollution exposure (Sullivan and Krupnick 2018; Fowlie et al. 2019; Chen et al. 2022).

¹¹ For more details about the availability of PM_{2.5} data for the GH's study period, see Appendix 4.

¹² For more details about spatiotemporal smoothing in MERRA-2 data, see Appendix 5.

¹³ Jin et al. (2022) conducted a comprehensive global-scale validation. Buchard et al. (2017) and Sayeed et al. (2022) evaluated MERRA-2 PM_{2.5} against ground-based PM_{2.5} in the United States. Similar efforts have been undertaken in mainland China and its subregions (Song et al. 2018; He et al. 2019; Ma et al. 2020; Ali et al. 2022). Studies have also focused on other parts of Asia, including Japan and Korea (Yin 2021), as well as Thailand (Gupta et al. 2021). Beyond Asia, assessments have extended to Africa (Nyasulu et al. 2023) and Turkey (Tuna Tuygun et al. 2022). Several studies have assessed the validity of MERRA-2 PM_{2.5} for India, including Navinya et al. (2020), Alang et al. (2022), and Dhandapani et al. (2023).

¹⁴ For more details about MERRA-2 validation studies and prediction errors, see Appendix 6.

¹⁵ Sources of SO₄ (sulfates), BC, and OC (carbonaceous) are emissions from power plants, vehicle exhaust, and biomass burning. Dust_{2.5} comes from local arid sources or transported from abroad by dust storms. SS_{2.5} penetrates the land from the seas and oceans.

¹⁶ For a detailed account of MERRA-2 PM_{2.5} construction, see Appendix 7.

¹⁷ For example, refer to the workflow used for deriving geospatial covariates in DHS datasets (DHS n.d.).

¹⁸ For a more detailed discussion on the availability of a "ground truth" for MERRA-2 PM_{2.5}, refer to Appendix 8.

¹⁹ Appendix 9 provides a detailed discussion of these findings and corroborates the observed rise in PM_{2.5}, reinforcing that these trends reflect actual environmental changes.

²⁰ The specification below represents the one-step version of the two-step approach. GH include both policies in the one-step model and restrict the policies' dummies to the observed event years—20 city years for CAT and 15 city years for SCAP—to maintain comparability with the two-step approach.

$$\begin{aligned}
 Y_{ct} = & \alpha + \theta_1 1(SCAP\ Range)_\tau + \theta_2 1(SCAP)_\tau \times (SCAP\ Range)_\tau \\
 & + \theta_3 1(SCAP\ Range)_\tau \times \tau \\
 & + \theta_4 1(SCAP)_\tau \times \tau \times (SCAP\ Range)_\tau + \theta_5 1(\tau Left)_\tau + \theta_6 1(\tau Right)_\tau \\
 & + \rho_1 1(CAT\ Range)_\phi \\
 & + \rho_2 1(CAT)_\phi \times (CAT\ Range)_\phi + \rho_3 1(CAT\ Range)_\phi \times \phi + \rho_4 1(CAT)_\phi \\
 & \times \phi \times (CAT\ Range)_\phi \\
 & + \rho_5 1(\phi Left)_\phi + \rho_6 1(\phi Right)_\phi + \beta X_{ct} + \mu_t + \gamma_c + \epsilon_{ct}
 \end{aligned}$$

$1(SCAP\ Range)_\tau$ is a dummy variable for $-7 \leq \tau \leq 3$, and $1(CAT\ Range)_\phi$ is a dummy variable for $-7 \leq \phi \leq 9$; $1(SCAP)_\tau$ and $1(CAT)_\phi$ are the policy dummies that indicate whether SCAP or CAT policies are in force and that take on the value 1 for the adopting cities with $\tau \geq 0$ and/or $\phi \geq 0$; $1(\tau Left)_\tau$ and $1(\tau Right)_\tau$ are dummies indicating that $\tau < -7$ or $\tau > 3$, respectively; by analogy, $1(\phi Left)_\phi$ and $1(\phi Right)_\phi$ indicate that $\phi < -7$ or $\phi > 9$, respectively; $1(SCAP\ Range)_\tau \times \tau$ and $1(CAT\ Range)_\phi \times \phi$ are a linear time trend variables interacted with a policy range dummies; $1(SCAP)_\tau \times \tau \times (SCAP\ Range)_\tau$ and $1(CAT)_\phi \times \phi \times (CAT\ Range)_\phi + \rho_5$ are policy $\times$ time-trend $\times$ policy-range interaction terms; ϵ_{ct} — standard errors clustered at the city-level.

²¹ It may well be that the coefficients on the SCAP policies' effects 5 years after implementation capture some other changes. Some blame lies with the energy generation by power plants, on which GH focus to a lesser degree than on vehicular pollution. Energy generation is the major contributor to air pollution in many developing countries and is certainly the driving force behind the rapid economic growth in China and India. At the city level, Goyal (2002) refers to the fossil fuel burning power plants in Delhi as the primary source of SO₂ and SPM air pollution, with the respective shares of 56.8% and 60.4%. For comparison, vehicular emissions contribute a modest 4.8% and 6.7% to SO₂ and SPM air pollution in Delhi. Thus, any increase in the power plant emissions increases levels of particulate air pollution. This can happen directly through the SPM channel and indirectly because of the conversion of SO₂ to sulfate particulates (SO₄), a PM_{2.5} component.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.